

Cover Letter -- IEEE TMM Submission

To: Editor-in-Chief, IEEE Transactions on Multimedia (TMM)

From: Jing An (joint first author and corresponding author) and Yanbing Bai (senior/PI corresponding author), on behalf of all co-authors

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Date: [INSERT SUBMISSION DATE]

Manuscript title: *An End-to-End Multimodal System for Subtitle Recognition and Chinese-Japanese Translation with Speech Synthesis in Short Dramas*

Dear Editor,

We are pleased to submit our manuscript, listed above, for consideration as a Regular Paper in *IEEE Transactions on Multimedia*. The paper presents what is, to the best of our knowledge, the first end-to-end multimodal localization system specifically designed for Chinese short-form drama (*duanju*), spanning the tightly coupled stages of visual subtitle recognition (frame-level OCR), audio transcription with adaptive multimodal fusion, neural machine translation, and zero-shot Japanese voice cloning for dubbed audio.

Why TMM is the appropriate venue

Short-form drama localization is fundamentally a multimedia problem: the subtitle text, the audio waveform, and the visual frame are jointly constrained, and progress on any single modality in isolation is insufficient for a deployable end-to-end pipeline. Our paper contributes (i) a confidence-adaptive OCR/ASR fusion algorithm that gates between the visual and audio channels using ASR log-probabilities; (ii) a domain-adapted vision-language translation component, with a controlled study of how visual context affects translation in zero-shot versus fine-tuned settings; (iii) a controlled comparison of four zero-shot Japanese voice cloning systems combining automatic intelligibility/similarity metrics with a blind human listening study; and (iv) a manually annotated 79-episode cross-modal benchmark with end-to-end error-propagation analysis and real-time-factor profiling. These contributions sit squarely within the TMM scope of multimodal signal processing, cross-modal analysis and fusion, multimedia systems, and applied multimodal machine learning, and the work spans vision, language, and speech in a way that is unlikely to find a natural home at any single-modality venue.

Relation to prior conference publication

This manuscript is a substantial extension of our preliminary conference paper accepted to **IEEE ICASSP 2026** (Session MMSP-P23), which introduced an early version of the OCR+ASR fusion idea and a small-scale LoRA fine-tuned Qwen2.5-3B translation model on a subset of the data. The present journal version extends that work along six independent dimensions, contributing well beyond the new-material expectation under the IEEE Policy on Prior Publication for journal extensions of conference papers. A separate **Summary of Differences** document detailing the extensions has been uploaded with this submission as required by IEEE policy.

1. **Comprehensive recognition benchmark:** eight OCR backbones evaluated under matched conditions (Qwen3-VL, Qwen2-VL, InternVL2, GOT-OCR2.0, Florence-2, RapidOCR, EasyOCR, TrOCR) versus the single Qwen2-VL configuration in the conference version.
2. **Frame-rate ablation:** a new study quantifying the 1 fps -> 5 fps recall gap from 27.7% -> 100%, absent from the conference paper.
3. **Upgraded fusion algorithm** (Algorithm 1): replaces the conference paper's fixed similarity threshold with a two-parameter confidence-adaptive gate using Whisper's segment log-probabilities, selected by grid search and formalized as new pseudocode.
4. **Cross-validated translation benchmark and fine-tuning scaling study:** LoRA fine-tuned **Qwen3-VL-4B** achieves 5-fold cross-validated **BLEU 15.70±4.42 / chrF++ 20.78±2.81 / COMET 0.843**, a +32% relative gain over its own zero-shot baseline (vs. BLEU 9.84 in the conference paper). We benchmark **eleven** baselines (incl. M2M-100, SeamlessM4T-v2, NLLB-200, InternVL3-8B, Qwen3.5-9B, Gemma-4-12B, Qwen3-8B) and fine-tune at multiple scales, with Gemma-4-12B reaching BLEU 30.75±1.84, showing domain fine-tuning helps across the 4B-12B range.
5. **Complete speech-synthesis stage:** four zero-shot TTS systems (GPT-SoVITS, CosyVoice2-0.5B, F5-TTS, EdgeTTS) compared with both automatic metrics and a **blind five-rater human listening study**, plus a reference-duration ablation; **entirely new** relative to the conference paper.
6. **New analyses:** visual-context translation ablation (+3.35 BLEU zero-shot; no benefit under fine-tuning), cross-stage error-propagation analysis, computational cost / real-time-factor profiling, and a 100-segment qualitative error taxonomy.

Originality and ethical compliance

- This manuscript is original work; it has not been previously published, and it is not under consideration at any other journal or conference.
- All listed co-authors have read and approved the submission and have agreed on author order.
- The dataset was provided by a commercial entertainment company under a research-use agreement (Section *Ethics Statement*). All performers consented to broadcast distribution; human annotators participated voluntarily and were compensated.
- We acknowledge the dual-use risk of zero-shot voice cloning and have chosen not to

release reference-audio-to-cloned-voice mappings; only short, content-neutral synthetic samples will be released for inspection (see Supplementary Materials).

- All authors declare **no competing financial or non-financial interests** related to this work.

Suggested reviewers (optional)

We do not propose specific reviewers, but we suggest the editorial board consider experts in any of the following overlapping areas: (i) video subtitling and multimedia localization, (ii) multimodal fusion for speech and video understanding, (iii) low-resource and domain-adapted neural machine translation for video content, and (iv) zero-shot voice cloning and TTS evaluation for video dubbing.

Reviewer-excluded list

We respectfully request that the following individuals not be invited as reviewers due to recent collaboration: *[INSERT NAMES IF ANY; LEAVE EMPTY OTHERWISE]*.

Compliance with TMM submission requirements

- **Format:** IEEE IEEEtran class, journal option (two-column, single-blind).
- **References:** IEEE Reference Format (IEEEtran.bst).
- **Keywords:** IEEE keywords block on the first page.
- **Summary of differences:** uploaded as a separate document for the journal extension of our ICASSP 2026 paper, per IEEE Policy on Prior Publication.
- **Supplementary materials:** README, configuration files, per-episode metric breakdowns, translation example pairs, and short anonymized synthesis samples are provided in the supplementary archive.
- **Reproducibility, Data Availability, and Ethics statements:** provided as separate sections at the end of the manuscript.

We thank you and the reviewers in advance for considering our submission and look forward to your feedback.

Sincerely,

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